MATHS MANTHAN

• TUITION CLASSES •

e-mail: contact@mathsmanthan.co.in | Website: mathsmanthan.co.in

Subject: Mathematics **WEEKLY CHAPTER TEST – 04 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 12: Surface Areas and Volumes** **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets ([**0.5 Mark**], [**1 Mark**], [**1.5 Marks**], [**2 Marks**]). Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions & Assertion-Reason MATHS MANTHAN | ($4 \times 1 = 4$ Marks)

Q1. If two solid hemispheres of the same base radius r are joined together along their bases, then the curved surface area of this new solid is: [1]

(A) $4\pi r^2$ (B) $6\pi r^2$ (C) $3\pi r^2$ (D) $8\pi r^2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: $\boxed{\textbf{(A) } 4\pi r^2}$ [0.5 Mark]

Reasoning: Joining two solid hemispheres of radius r base-to-base forms a complete solid sphere of radius r (the flat bases become internal).

Formula / Theorem Used: Curved Surface Area of New Solid (Sphere) $= 2\pi r^2 + 2\pi r^2 = \boxed{\mathbf{4\pi r^2}}$

Q2. A spherical ball of radius r is melted to make 8 new identical balls each of radius r_1. Then r_1 is equal to: [1]

(A) r (B) $\frac{r}{2}$ (C) $\frac{r}{4}$ (D) $2r$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: $\boxed{\textbf{(B) } \frac{r}{2}}$ [0.5 Mark]

Given: Volume of big sphere $= 8 \times$ Volume of one small ball.

Formula / Theorem Used: $\frac{4}{3}\pi r^3 = 8 \times \frac{4}{3}\pi r_1^3$

Cancelling $\frac{4}{3}\pi$: $r^3 = 8r_1^3 = (2r_1)^3 \implies 2r_1 = r \implies \boxed{\mathbf{r_1 = \frac{r}{2}}}$.

Q3. Water flows at the rate of 10 metres per minute through a cylindrical pipe of internal diameter 5 mm. How long will it take to fill a conical vessel of base diameter 40 cm and depth 24 cm? [1]

(A) 48 min 15 s (B) 51 min 12 s (C) 52 min 10 s (D) 53 min

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 51 min 12 s** [0.5 Mark]

Given: Pipe: $r_1 = 2.5$ mm $= 0.25$ cm $= \frac{1}{4}$ cm, speed $v = 10$ m/min $= 1000$ cm/min. Cone: $r_2 = 20$ cm, $h_2 = 24$ cm.

Formula / Theorem Used: Time $(t) = \dfrac{\text{Volume of Cone}}{\text{Water Volume per Minute}} = \dfrac{\frac{1}{3}\pi r_2^2 h_2}{\pi r_1^2 v}$

$$t = \frac{\frac{1}{3} \times 400 \times 24}{\left(\frac{1}{4}\right)^2 \times 1000} = \frac{3200}{\frac{1000}{16}} = \frac{3200 \times 16}{1000} = \frac{256}{5} = 51.2 \text{ min} = 51 \text{ min } (0.2 \times 60) \text{ s} = \boxed{\mathbf{51} \text{ min } \mathbf{12} \text{ s}}$$

Q4. **Assertion (A):** If the surface areas of two spheres are in the ratio 16 : 9, then their volumes are in the ratio 64 : 27.

Reason (R): If S_1, S_2 are surface areas of two spheres and V_1, V_2 are their volumes, then $\dfrac{V_1}{V_2} = \left(\dfrac{S_1}{S_2}\right)^{3/2}$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** [0.5 Mark]

Verification: Since $S \propto r^2$ and $V \propto r^3$, $\frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\sqrt{\frac{S_1}{S_2}}\right)^3 = \left(\frac{S_1}{S_2}\right)^{3/2}$ (Reason (R) is true). [0.5 Mark]

Substituting $\frac{S_1}{S_2} = \frac{16}{9}$: $\frac{V_1}{V_2} = \left(\frac{4}{3}\right)^3 = \frac{64}{27}$. Both (A) and (R) are true, and (R) explains (A).

SECTION B: Short Answer Questions

MATHS MANTHAN | (2 × 2 = 4 Marks)

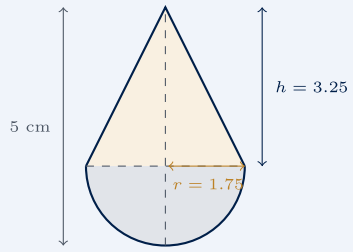
Q5. Rasheed got a playing top (lattu) as his birthday present, which had no colour on it. The top is shaped like a cone surmounted by a hemisphere. The entire top is 5 cm in height and diameter is 3.5 cm. Find the area he has to colour. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Dimensions & Slant Height [1 Mark]

Radius $r = \frac{3.5}{2} = 1.75$ cm; cone height $h = 5 - 1.75 = 3.25$ cm.

$l = \sqrt{r^2 + h^2} = \sqrt{(1.75)^2 + (3.25)^2} = \sqrt{3.0625 + 10.5625} = \sqrt{13.625} \approx 3.69$ cm.



Formula / Theorem Used: Colour Area $= \text{CSA}_{\text{cone}} + \text{CSA}_{\text{hemi}} = \pi r(l + 2r)$

Step 2: Substitution $(\pi = \frac{22}{7})$ [1 Mark]

$$\text{Area} = \frac{22}{7} \times \frac{3.5}{2} \times (3.691 + 3.5) = 5.5 \times 7.191$$

$$\approx \boxed{\mathbf{39.55} \text{ cm}^2} \text{ (or } 39.6 \text{ cm}^2 \text{ using } l \approx 3.7 \text{ cm)}$$

Q6. A decorative block is made of two solids — a cube and a hemisphere. The base is a cube with edge 5 cm, and the hemisphere fixed on top has diameter 4.2 cm. Find the total surface area of the block. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Declaration [1 Mark]

Cube edge $a = 5$ cm; hemisphere diameter $d = 4.2$ cm $\implies$ $r = 2.1$ cm.

Formula / Theorem
Used: TSA of Block $= \text{TSA}_{\text{cube}} - \text{Base Area}_{\text{hemi}} + \text{CSA}_{\text{hemi}} = 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$

Step 2: Arithmetic [1 Mark]

$6a^2 = 6 \times 5^2 = 150$ cm^2; $\quad \pi r^2 = \frac{22}{7} \times 2.1 \times 2.1 = 22 \times 0.3 \times 2.1 = 13.86$ cm^2.

$$\text{TSA} = 150 + 13.86 = \boxed{\mathbf{163.86}\ \text{cm}^2}$$

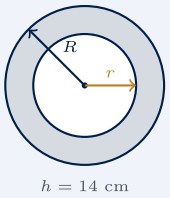
SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q7. The difference between the outside and inside curved surface areas of a cylindrical metallic pipe 14 cm long is 44 cm^2. If the pipe is made of 99 cm^3 of metal, find the outer and inner radii of the pipe. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Using Difference of Curved Surface Areas [1 Mark]

Let outer radius $= R$, inner radius $= r$, length $h = 14$ cm.



Formula / Theorem Used: Diff. of CSA $= 2\pi(R-r)h$ and Volume of Metal $= \pi(R^2 - r^2)h$

$$2 \times \frac{22}{7} \times (R-r) \times 14 = 44 \implies 88(R-r) = 44$$

$$R - r = \frac{1}{2} = 0.5 \text{ cm} \quad \text{— (i)}$$

Step 2: Using Volume of Metal [1 Mark]

$$\frac{22}{7} \times (R^2 - r^2) \times 14 = 99 \implies 44(R-r)(R+r) = 99$$

Substituting $R - r = \frac{1}{2}$: $44\left(\frac{1}{2}\right)(R+r) = 99 \implies 22(R+r) = 99$

$R + r = \frac{9}{2} = 4.5$ cm — (ii)

Step 3: Solving (i) and (ii) [1 Mark]

Adding (i) and (ii): $2R = 5 \implies R = 2.5$ cm; $\quad r = 4.5 - 2.5 = 2$ cm.

$\boxed{\mathbf{R = 2.5}\ \text{cm},\ \mathbf{r = 2}\ \text{cm}}$

Q8. A wall of length 24 m, thickness 0.4 m, and height 6 m is constructed with bricks, each of dimensions 25 cm × 16 cm × 10 cm. If mortar occupies $\frac{1}{10}$th of the total volume of the wall, find the number of bricks used. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Converting Units & Wall Volume [1 Mark]

Wall dimensions in cm: $L = 2400$ cm, $B = 40$ cm, $H = 600$ cm.

$V_{\text{wall}} = 2400 \times 40 \times 600$ cm^3. Volume occupied by bricks $= \left(1 - \frac{1}{10}\right) V_{\text{wall}} = \frac{9}{10} V_{\text{wall}}$.

Formula / Theorem Used: Number of Bricks $(n) = \dfrac{\frac{9}{10} \times \text{Volume of Wall}}{\text{Volume of One Brick}}$

Step 2: Substitution & Cancellation [2 Marks]

$$n = \frac{\frac{9}{10} \times 2400 \times 40 \times 600}{25 \times 16 \times 10} = \frac{9 \times 240 \times 40 \times 600}{4000} = 9 \times 240 \times 6 = \boxed{\mathbf{12960}\ \text{bricks}}$$

Q9. Spherical marbles of diameter 1.4 cm are dropped into a cylindrical beaker of diameter 7 cm containing some water.
(i) Find the volume of water displaced when the water level rises by 5.6 cm.
(ii) Calculate the volume of one spherical marble.
(iii) Determine the number of marbles that must be dropped to achieve this rise in water level. [6]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 6 Marks]

Given: Beaker radius $R = \frac{7}{2} = 3.5$ cm, water rise $h = 5.6$ cm. Marble radius $r = \frac{1.4}{2} = 0.7$ cm.

Formula / Theorem Used: $V_{\text{displaced}} = \pi R^2 h$ and $V_{\text{marble}} = \frac{4}{3}\pi r^3$

Step 1: (i) Volume of Water Displaced [2 Marks]

$$V_{\text{displaced}} = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 5.6 = 11 \times 7 \times 2.8 = \boxed{\mathbf{215.6}\ \text{cm}^3}$$

Step 2: (ii) Volume of One Spherical Marble [2 Marks]

$$V_{\text{marble}} = \frac{4}{3} \times \frac{22}{7} \times \frac{7}{10} \times \frac{7}{10} \times \frac{7}{10} = \frac{4 \times 22 \times 49}{3 \times 1000} = \frac{4312}{3000} = \frac{539}{375} = \boxed{\approx \mathbf{1.437}\ \text{cm}^3}$$

Step 3: (iii) Number of Marbles (n) [2 Marks]

$$n = \frac{V_{\text{displaced}}}{V_{\text{marble}}} = \frac{215.6}{\frac{539}{375}} = \frac{1078}{5} \times \frac{375}{539} = 2 \times 75 = \boxed{\mathbf{150}\ \text{marbles}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★